AN ETALE EQUIVALENCE RELATION ON A SCHEME PRESENTS AN ALGEBRAIC SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. This candidate was dropped by the mechanical prefilter because of the phrase 'It is clear that coproduct F_i -> F is surjective' at proof line 1853; the read shows a one-line observation inside a three-reduction complete proof.

Editorial note. The references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Allowed standard prerequisites (editorial).

- The quotient is unchanged by restricting along a surjective etale morphism. Groupoids, Definition groupoids-definition-restrict-relation and Lemma groupoids-lemma-quotient-groupoid-restrict, used at proof lines 1833-1837. Allows replacing U by U'.
- Restriction of an etale equivalence relation is again one. Lemma lemma-pullback-etale-equivalence-relation (1500-1510), used at proof lines 1837-1838 and 1847-1849. Keeps the hypotheses after restriction.
- The affine case of the presentation theorem. Lemma lemma-presentation-quasi-compact (1682-1690, produced as L44199), used at proof lines 1853-1855. Makes each F_i an algebraic space.
- The quotient map of an etale equivalence relation is etale and surjective. Lemma lemma-when-it-works-it-works (1647-1656, produced as L44180), used at proof lines 1829-1831 and 1855-1857. Reduces the claim to algebraicity of U/R and gives the properties of U_i -> F_i.
- A coproduct of algebraic spaces is an algebraic space. Lemma lemma-coproduct-algebraic-spaces (1276-1285), used at proof lines 1857-1858. Makes coproduct F_i an algebraic space.
- Gluing a sheaf from open algebraic subfunctors. Lemma lemma-glueing-algebraic-spaces (1298-1318, counted as L44174), used at proof lines 1858-1861. Concludes that F = U/R is an algebraic space.

Source reference. Original definition 02V9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3760–3760. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02VH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3768–3768. Navigation pointer into the Stacks

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Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3813–3813. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3817–3817. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3814–3814. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0265: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2838–2838. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3816–3816. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3818–3818. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Theorem 0.2. *Let S be a scheme. Let U be a scheme over S. Let $j = (s, t) : R \to U \times_S U$ be an étale equivalence relation on U over S. Then the quotient U/R is an algebraic space, and $U \to U/R$ is étale and surjective, in other words $(U, R, U \to U/R)$ is a presentation of U/R.*

Proof. By Lemma 02WV it suffices to prove that U/R is an algebraic space. Let $U' \to U$ be a surjective, étale morphism. Then $\{U' \to U\}$ is in particular an fppf covering. Let R' be the restriction of R to U', see Groupoids, Definition 02V9. According to Groupoids, Lemma 02VH we see that $U/R \cong U'/R'$. By Lemma 02WT R' is an étale equivalence relation on U'. Thus we may replace U by U'.

We apply the previous remark to $U' = \coprod U_i$, where $U = \bigcup U_i$ is an affine open covering of U. Hence we may and do assume that $U = \coprod U_i$ where each U_i is an affine scheme.

Consider the restriction R_i of R to U_i. By Lemma 02WT this is an étale equivalence relation. Set $F_i = U_i/R_i$ and $F = U/R$. It is clear that $\coprod F_i \to F$ is surjective. By Lemma 02WU each $F_i \to F$ is representable, and an open immersion. By Lemma 0265 applied to (U_i, R_i) we see that F_i is an algebraic space. Then by Lemma 02WV we see that $U_i \to F_i$ is étale and surjective. From Lemma 02WQ it follows that $\coprod F_i$ is an algebraic space. Finally, we have verified all hypotheses of Lemma 02WR and it follows that $F = U/R$ is an algebraic space. □